The ordering-odd inner product
A relative Casimir and its translation test

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Abstract

The known leading commutator four-point function determines a positive two-particle quadratic form: the difference between the total and the two one-particle Casimirs. We derive this directly from coherent-state matrix elements, without assuming the conjectured all-degree finite-q ordering overlap. For the quadratic and cubic singlets the relative eigenvalues are $k(k+9)+12$. We construct self-adjoint light-front translations in this weighted space and then test the stronger condition that they preserve the microscopic product vertex. An explicit norm calculation distinguishes these two requirements.

1 The leading inner product

Use the annealed normalized trace of complex Gaussian p-body DSSYK, standard CAR including overlapping index sets, and $\beta = \mu_{\rm chem} = 0$. First take $N, p \to \infty$ at $\lambda = p^2/N > 0$, then small $h = 4\lambda$, with $q = e^{-h}$, $u = 2\mathcal{J}_s t$ and $\hbar = 1$. The normalized derivative singlets of the preceding notes have labels $d = 2, e = 3$ [4].

The free limiting product representation is $\mathcal{H}_0 = D_d^+ \otimes D_e^+$. Write its generators as E_j, B_j, Π_j, with $C_j = E_j^2 - B_j^2 - \Pi_j^2 = d_j(d_j-1)$ and primitive total generators. The leading norm of the rescaled commutator is

$$\boxed{\langle f, g\rangle_{\rm odd} = \langle f, \mathcal{A} g\rangle_0, \qquad \mathcal{A} = C_{\rm tot} - C_1 - C_2 = 2(E_1E_2 - B_1B_2 - \Pi_1\Pi_2).} \tag{1}$$

For $d = 2, e = 3$, the relative-channel spectrum is

$$\boxed{\mathcal{A}_k = k(k+9) + 12, \qquad k = 0, 1, \dots.} \tag{2}$$

This is the entire positive quadratic form of the leading theory, not a finite-q all-k overlap formula. It identifies the norm required by the extra ordering channel before a spatial dynamics is chosen.

Lin–Stanford already give the leading commutator kernel and its block weights [1]. We establish their operator realization for the present singlets and use it to examine translations. No priority claim for the commutator sector or the discrete-series decomposition is made.

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2 Direct identification from the known kernel

Realize D_d^+ on $L^2(\mathbb{R}_+, dp)$ with

$$Q = p, \qquad B = i(p\partial_p + \tfrac{1}{2}), \qquad S = -p\partial_p^2 - \partial_p + (d-\tfrac{1}{2})^2/p,$$
$$E = (Q+S)/2, \qquad \Pi = (Q-S)/2, \qquad \psi_d(p) = \frac{2^d}{\sqrt{\Gamma(2d)}} p^{d-1/2} e^{-p}. \tag{3}$$

The states $|u\rangle = e^{-2iuB}\psi_d$ have real wavefunctions $e^u \psi_d(e^{2u}p)$ and overlap $g_d(u-u') = \operatorname{sech}^{2d}(u-u')$. Elementary gamma integrals give

$$\frac{\langle u'|E|u\rangle}{g_d(u-u')} = d\frac{\cosh(u+u')}{\cosh(u-u')}, \qquad \frac{\langle u'|B|u\rangle}{g_d(u-u')} = -id\tanh(u-u'),$$
$$\frac{\langle u'|\Pi|u\rangle}{g_d(u-u')} = -d\frac{\sinh(u+u')}{\cosh(u-u')}. \tag{4}$$

Taking tensor products and substituting in (1) yields

$$\langle u', v'|\mathcal{A}|u, v\rangle = 2de\, \operatorname{sech}^{2d+1}(u-u')\operatorname{sech}^{2e+1}(v-v') \times \big[\cosh(u+u'-v-v') + \sinh(u-u')\sinh(v-v')\big]. \tag{5}$$

This reproduces the covariance derived in the preceding note from Lin–Stanford Eqs. (170), (205) and (388):

$$B_h(u,v) = -i[V(u), W(v)]/\sqrt{h}, \qquad \lim_{h\to 0} \tau(B_h(u',v')B_h(u,v)) = \langle u', v'|\mathcal{A}|u, v\rangle.$$

The equality includes arbitrary finite linear combinations of time insertions. It uses the sourced leading fixed-time kernel, and does not require differentiating a numerically fitted function.

Closure and completeness. The tensor decomposition [2] reads $D_d^+ \otimes D_e^+ = \bigoplus_{k\geq 0} D_{d+e+k}^+$. Thus $\mathcal{A}_k = k(k+2d+2e-1)+2de \geq 2de > 0$. Its spectral realization defines a positive self-adjoint operator. Finite tensor descendant sums are a form core. Derivatives at zero of the boost orbits span these descendants: $B^n\psi_d$ is triangular in the ladder basis with a nonzero top coefficient. The analytic orbit vectors admit these difference quotients also in the form norm. Consequently their span is form-dense, and the kernel identifies the closed leading form uniquely. This is not a claim of convergence uniform in channel number or in times growing with h^{-1}.

3 An explicit relative differential operator

Use $P = p_1 + p_2 > 0$, $x = p_1/P \in (0,1)$. After extracting the square root of the beta weight, the product Hilbert space is

$$\mathcal{H}_0 = L^2(dP\, w(x)dx), \qquad w(x) = \frac{x^{2d-1}(1-x)^{2e-1}}{\mathrm{B}(2d,2e)}. \tag{6}$$

The Jacobi–Laguerre tensor map [2, 4] gives

$$\mathcal{A} = -x(1-x)\partial_x^2 - [2d - 2(d+e)x]\partial_x + 2de. \tag{7}$$

Its closed quadratic form, initially on finite Jacobi sums, is

$$\|f\|^2_{\mathrm{odd}} = \int dP \int_0^1 w(x) \left[x(1-x)|\partial_x f|^2 + 2de|f|^2\right] dx. \tag{8}$$

The normalized relative eigenfunctions are Jacobi polynomials $p_k(x) \propto P_k^{(2e-1,2d-1)}(2x-1)$. At $d = 2, e = 3$,

$$w(x) = 504x^3(1-x)^5, \quad \mathcal{A} = -x(1-x)\partial_x^2 - (4-10x)\partial_x + 12,$$

and its first six eigenvalues are $12, 22, 34, 48, 64, 82$. The equal-time normalized commutator of the last note is the $k = 0$ seed, divided by $\sqrt{12}$ in this form norm.

Define $\mathcal{H}_{\mathrm{odd}} = D(\mathcal{A}^{1/2})$, completed in (8). Since $\mathcal{A} \geq 12$, it is already complete in that norm, and

$$U_{\mathcal{A}} : \mathcal{H}_{\mathrm{odd}} \to \mathcal{H}_0, \qquad U_{\mathcal{A}} f = \mathcal{A}^{1/2} f \tag{9}$$

is unitary and onto. Its inverse is bounded on $\mathcal{H}_0$. The total generators E, B, Q commute strongly with $\mathcal{A}$ in the direct-sum representation. Their actions therefore carry over unchanged to the weighted space; specifically $Q = P$ and $B = i(P\partial_P + \frac{1}{2})$.

The product vertex in normalized coordinates. Writing an ordered product as half the anticommutator plus half the commutator, the leading map into the normalized odd copy is

$$\mathcal{V}_h f = \frac{i\sqrt{h}}{2} \mathcal{A}^{1/2} f. \tag{10}$$

On $k = 0$ it is $i\sqrt{3h} = i\sqrt{12\lambda}$, reproducing the previous three-point overlap. The amplitude grows with relative k; the fixed-channel expansion is not uniform for hk^2 of order one. The closed form specifies which wave packets have finite leading production norm: exactly $f \in D(\mathcal{A}^{1/2})$.

4 A norm-preserving translation construction

For positive constituent masses m_1, m_2 the standard free light-front mass profile [3] is

$$M(x) = \frac{m_1^2}{x} + \frac{m_2^2}{1-x}, \qquad P^+ = \mu P, \qquad P_0^- = \frac{M(x)}{\mu P}, \qquad \mu > 0. \tag{11}$$

Here P, x are dimensionless, whereas m_j, μ have energy units. Multiplication by M is positive self-adjoint in $\mathcal{H}_0$, with spectrum $[(m_1+m_2)^2, \infty)$. It cannot simply be reused in the odd metric:

$$\begin{aligned}[\mathcal{A}, M]f = & -2x(1-x)M'f' \\ & -\big[x(1-x)M'' + (4-10x)M'\big]f.\end{aligned} \tag{12}$$

This is nonzero for the free profile. The coefficient of f' shows that a smooth multiplication profile commuting with $\mathcal{A}$ must be constant.

There is an explicit self-adjoint construction in the weighted space:

$$\boxed{\mathcal{M}_{\rm odd}^2 = \mathcal{A}^{-1/2} M \mathcal{A}^{1/2}, \qquad P_{\rm odd}^- = \frac{\mathcal{M}_{\rm odd}^2}{\mu P}.} \tag{13}$$

Its domain is the inverse image under $U_{\mathcal{A}}$ of the multiplication domain of M (and of $M/(\mu P)$ for P^-). Unitary transport proves self-adjointness, positivity, and the same mass spectrum. It also gives strongly commuting P^+, P^- and

$$[B, P^+] = iP^+, \qquad [B, P^-] = -iP^-, \qquad [P^+, P^-] = 0$$

on transported smooth cores. This solves the Hilbert-space consistency of a candidate free translation; it does not yet match the product vertex.

An exact diagnostic. Set $m_1 = m_2 = 1$ in a fixed-P fiber, and use $p_0 = 1$, $p_1 = (10x - 4)/\sqrt{24/11}$. Then $\langle p_0, Mp_0\rangle_0 = 24/5$ and $\langle p_0, Mp_1\rangle_0 = -\sqrt{66}/10$. For $f = (p_0 + ip_1)/\sqrt{2}$, direct integration gives

$$\boxed{\|e^{-isM} f\|_{\rm odd}^2 = 17 + \sqrt{66}\, s + 181 s^2.} \tag{14}$$

Thus the undressed multiplication phase genuinely fails to preserve the physical odd norm; this is not a finite matrix truncation effect. The transported evolution (13) preserves it exactly.

5 The stronger product-compatibility condition

Let $T = \mathcal{A}^{1/2}$, so the microscopic leading vertex in normalized coordinates is $i\sqrt{h}\,T/2$. Suppose it is to be a translation intertwiner between an ordinary two-particle mass M and some self-adjoint odd-sector mass N:

$$e^{-isN}Tf = Te^{-isM}f, \qquad f \in D(T), \tag{15}$$

with domain invariance. Taking norms forces $\|Te^{-isM}f\|_0 = \|Tf\|_0$, i.e. invariance of the $\mathcal{A}$ form under the even-sector evolution. Equation (14) contradicts this for the primitive free profile. No choice of self-adjoint N on this closed target sector can fix that particular intertwining requirement.

Equivalently, on a suitable common core the necessary condition is $[\mathcal{A}, M] = 0$. In normalized relative coordinates,

$$[\mathcal{A}, M]_{01} = \sqrt{66}, \qquad [T, M]_{01} = (\sqrt{12} - \sqrt{22})(-\sqrt{66}/10) \neq 0. \tag{16}$$

The transported odd mass on page 4 becomes simply M after applying $U_{\mathcal{A}}$. It therefore preserves norms but fails (15). The two statements have different requirements and must not be conflated.

What can satisfy both conditions? A positive relative mass operator $F(\mathcal{A})$ commutes with the vertex, and yields kinematically consistent translations and intertwining on both copies. Its masses $F(k(k+9)+12)$ remain arbitrary; choosing such a function is not a derivation of meson dynamics. More generally the assumptions behind (15) may have to change: the microscopic product insertion need not already be an on-shell transition map, and physical translations may involve additional sectors or a different spatial operator dictionary. The conditional incompatibility concerns the specified free-pair identification, not the existence of a bulk dual.

6 Checks of the actual claims

Fifty exact symbolic identities verify the coherent Casimir kernel, general differential commutator, Jacobi conjugation, six relative eigenfunctions, their full orthogonality matrix, and the translation diagnostic. The 6×6 free mass matrix is retained. Five independent relative-coordinate integrals at 55 and 85 digits reproduce the four-time kernel; the largest 85-digit absolute residual is $5.15\,10^{-85}$. Four 70-digit integrals check (14), with residual below $2.90\,10^{-70}$. These assess numerical precision; the displayed identities also have analytic derivations.

A Finite-h controls and the next physical input

Independent finite-q direct/cross Gram recursions construct primaries through $k = 4$ for $d = 2, e = 3$, at $h = .04, .02, .01$. For the parity-adjusted overlap ρ_k, compare $2(1 - \rho_k)/h$ with (2). At $h = .01$ the results are

k	finite-h odd norm divided by h	leading eigenvalue
0	11.64709328	12
1	20.83317294	22
2	31.26703668	34
3	42.67442779	48
4	54.77019259	64

Increasing precision from 85 to 115 digits changes the coefficients relatively by at most $4.37\,10^{-79}$. Nevertheless the finite-h deviation at $k = 4$ is still 14.42%. Numerical precision does not remove the finite-deformation correction.

Lin–Stanford Appendix F describes its general finite-q overlap formula, Eq. (355), as inferred from small-degree calculations. Our all-k leading form is instead obtained from the independently available leading commutator kernel, its coherent-state realization, and the standard tensor Casimir. The low-degree finite-q comparisons are checks, not the proof of that limiting operator. We make no uniform large-k or finite-h operator-norm convergence assertion.

Next research step. The leading ordering metric and its domain are now explicit. A free translation can be made unitary, but its compatibility with the microscopic multiplication data requires extra physics. The next step is to return to the Susskind bulk gauge/meson construction and identify which spatial insertions or interacting translation generators its dictionary actually proposes. The tests are the norm (8), the vertex (10), and the conditional Ward relation (15). We should derive or test that map, rather than choose an arbitrary function of the Casimir and call it a mass spectrum.

References

[1] H. W. Lin and D. Stanford, *A symmetry algebra in double-scaled SYK*, arXiv:2307.15725, Section 4.3, Eq. (170), and Appendix F.1.

[2] H. T. Koelink and J. Van der Jeugt, *Convolutions for orthogonal polynomials from Lie and quantum algebra representations*, q-alg/9607010, Section 3, Eq. (3.2), Corollary 3.6 and Remark 3.7.

[3] H.-C. Pauli, *A compendium of light-cone quantization*, hep-ph/0103106, Eq. (87).

[4] A. Day and Chengheng, local notes *Two matter*, *Antisymmetric descendants*, and *Commutator primary* (November 2026). Evidence: `results/susskind-ordering-metric`.